MIL-T-26772F(USAF)
19 MAY 1970
SUPERSEDING
MIL-T-26772E(USAF)
17 September 1963

MILITARY SPECIFICATION

TRAILER, COMPRESSED GAS CYLINDER, AIR OR NITROGEN, OIL PUMPED, SERVICING, 2-WHEEL, 8-CYLINDER, 3500 PSI, TYPE MD-3A

1. SCOPE

1.1 This specification covers one type of air or nitrogen servicing, compressed gas cylinder trailer, designated Type MD-3A.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

RR-C-901	Cylinders, Steel, Seamless, Type 3A (For Compressed Gases)
UU-T-81	Tags, Shipping and Stock
PPP-C-650	Crate, Wood, Open and covered
PPP-T-60	Tape, Pressure Sensitive Adhesive, Waterproof - For Packaging and Sealing

Military

MIL-T-3065	Rubber, Fabricated Parts
MIL-V-4719	Valve, Oxygen Line, High Pressure, Aircraft Servicing
MIL-C-4810	Cylinder, Oxygen Purifier, Type MB-1
MIL-I-6866	Inspection, Penetrant Method of
MIL-I-6868	Inspection Process, Magnetic Particle
MIL-T-7003	Trichlorethylene, Stabilized Degreasing
MIL-W-8005	Wheels and Hubs for Industrial Pneumatic Tires
MIL-M-8090	Mobility Requirements, Ground Support Equipment, General Specification for
MIL-A-8421	Air Transportability Requirements, General Specification for
MIL-R-13689	Reflectorized Sheeting, Adhesive (Retro-Reflective)
MIL-B-22182	Brakes, Parking, Mobile Ground Support Equipment
MIL-M-26512	Maintainability Requirements for Aerospace Systems and Equipment

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:12Z
Check the source to verify that this is the current version before use.

MIL-H-26666	Hose Assembly, Pneumatic, High Pressure
MIL-T-27730	Tape, Antiseize, Tetrafluoroethylene, with Dispenser
MIL-R-38489	Reels, Static Discharge, Grounding, RMU-17/E and RMU-18/E

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-808	Finishes, Protective, and Codes, for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking and Waterproofing with Appropriate Test Methods
MS33656	Fitting End, Standard Dimensions for Flared Tube Connection and Gasket Seal

Air Force-Navy Aeronautical

AN6235	Filter Element - Hydraulic Replacement Micronic Line Type

DRAWINGS

Air Force

44A19817	Chuck Assembly - High Pressure Air
48B7796	Ring Assembly - Tie Down, 10,000 Lb.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Interstate Commerce Commission

49CFR 71-78	Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (1949 Edition-Revised 1950) available from the Superintendent of Documents, Government Printing Office, Washington 25, D.C. Orders for the above publication should cite "49 CFR 71-78 (Rev. 1950.)".

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:12Z
Check the source to verify that this is the current version before use.

Compressed Gas Association

Pamphlet S-1 Safety Relief Device Standards

(Application for copies should be addressed to the Compressed Gas Association, Incorporated, 11 West 42nd Street, New York 18, New York.)

American Society of Mechanical Engineers Power Test Code

19.5:4 Supplements on Instruments and Apparatus; Part 5, Chapter 4, Flow Measurement by Means of Standardized Nozzles and Orifice Plates

(Application for copies should be addressed to the American Society of Mechanical Engineers, 29 West 39th Street, New York 18, New York.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Components. The trailer shall consist of the following major components (see figures 1, 2, and 3):

DESCRIPTION	SEE REQUIREMENT
Chassis	3.8.1
Storage cylinders	3.8.2
Manifold system	3.8.3
Servicing system	3.8.4
Control panel	3.8.5
Utility and storage box	3.8.6

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials.

3.4.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be compounded or treated as necessary to insure against fungus attack. Fungicidal treatment will not be necessary for materials used in a hermetically sealed enclosure.

3.4.2 Metals. Wherever practicable, lightweight metals shall be used in the construction of the trailer.

3.4.3 Rubber goods. All rubber goods, such as seals, shall be fabricated from rubber compounds conforming to MIL-R-3065.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:12Z
Check the source to verify that this is the current version before use.

3.5 General design. The trailer shall be a readily mobile gas storage and servicing unit designed for:

a. Being charged with compressed air or nitrogen at a supply location or having cylinders removed and replaced.

b. Transporting the compressed air or nitrogen on an airbase or similar installation to the aircraft or missile equipment in need of servicing.

c. Charging the aircraft or missile equipment to the required pressure with the contained compressed air or nitrogen.

3.5.1 Configuration. The equipment overall design configuration shall be that of a 2-wheel trailer with a retractable landing wheel.

3.5.2 Reliability. The trailer shall have a mean-time-between-failure (MTBF) of not less than 21 air servicing to 3,000 psi cycles with a minimum reliability of 95 percent for a 1-cycle mission at a confidence factor of 0.90. (see 6.3.1 and 6.3.2).

3.5.3 Maintainability. The trailer design and construction shall provide for compliance with the maintainability requirements specified in MIL-M-26512.

3.5.3.1 To the maximum practicable extent, maintainability provisions shall incorporate features insuring operating clearance for facilitating maintenance and servicing at low ambient temperatures by personnel wearing heavy gloves or mittens and handicapped by bulky clothing and footgear.

3.5.3.2 Intricate locking devices, controls and threaded fastenings that can be easily overtorqued by personnel lacking feel through thick gloves or numbness shall be avoided where possible.

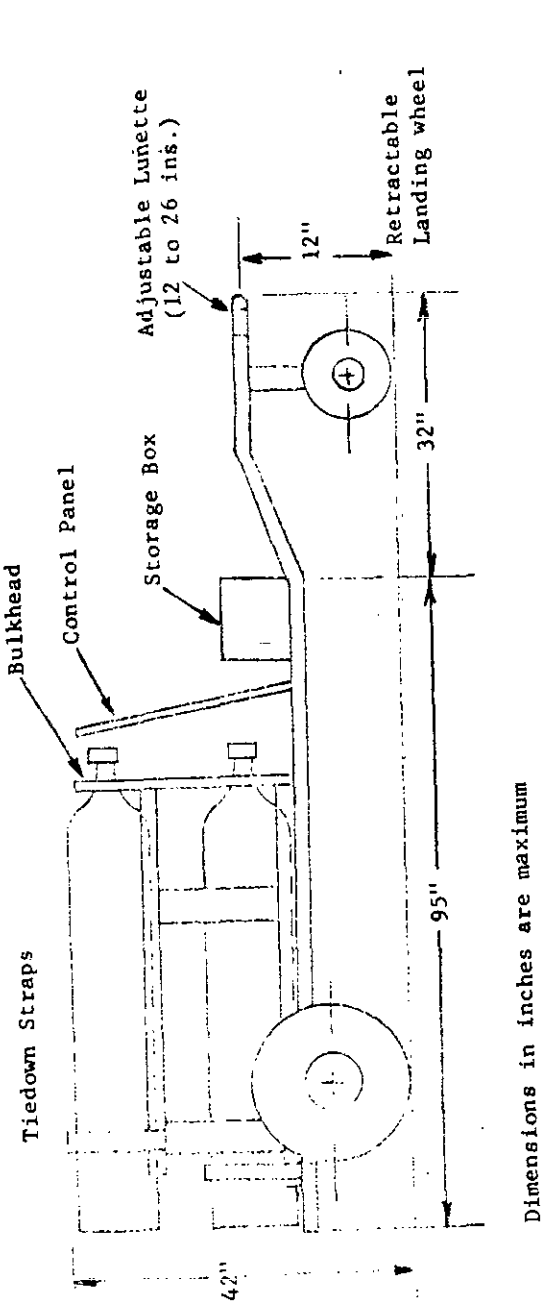
3.5.3.3 Covers or plates that must be removed for component adjustments or for component or part removal shall be equipped with substantial quick-disconnect fastenings.

3.5.4 Fool proofness. Where improper installation of an item could cause malfunction of the item or the equipment in which it is installed, an unsymmetrical mounting means shall be provided. This mounting shall be so designed that the item can be installed only in its proper operating position.

3.5.5 Lubrication. Wherever practicable, lubricated-for-life components or components not requiring lubrication shall be used in the trailer design. Grease and oil seals shall be designed and located to provide maximum accessibility for inspection and replacement.

3.5.5.1 Parts subject to wear. The trailer shall be designed to provide for lubrication of parts subject to wear. Such parts shall be designed and constructed to be readily renewable. Grease fittings and other lubrication points shall be readily accessible.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:12Z
Check the source to verify that this is the current version before use.



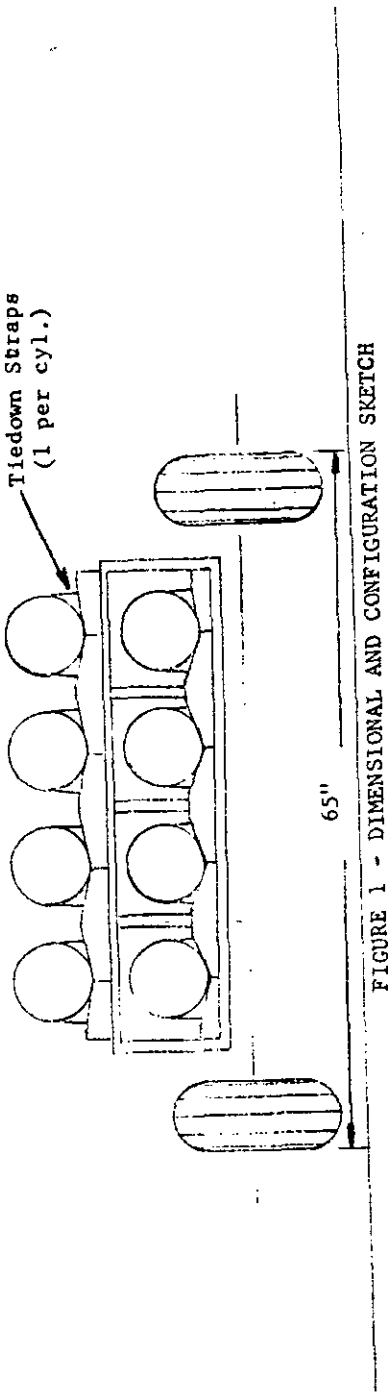


FIGURE 1 - DIMENSIONAL AND CONFIGURATION SKETCH

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:12Z
Check the source to verify that this is the current version before use.

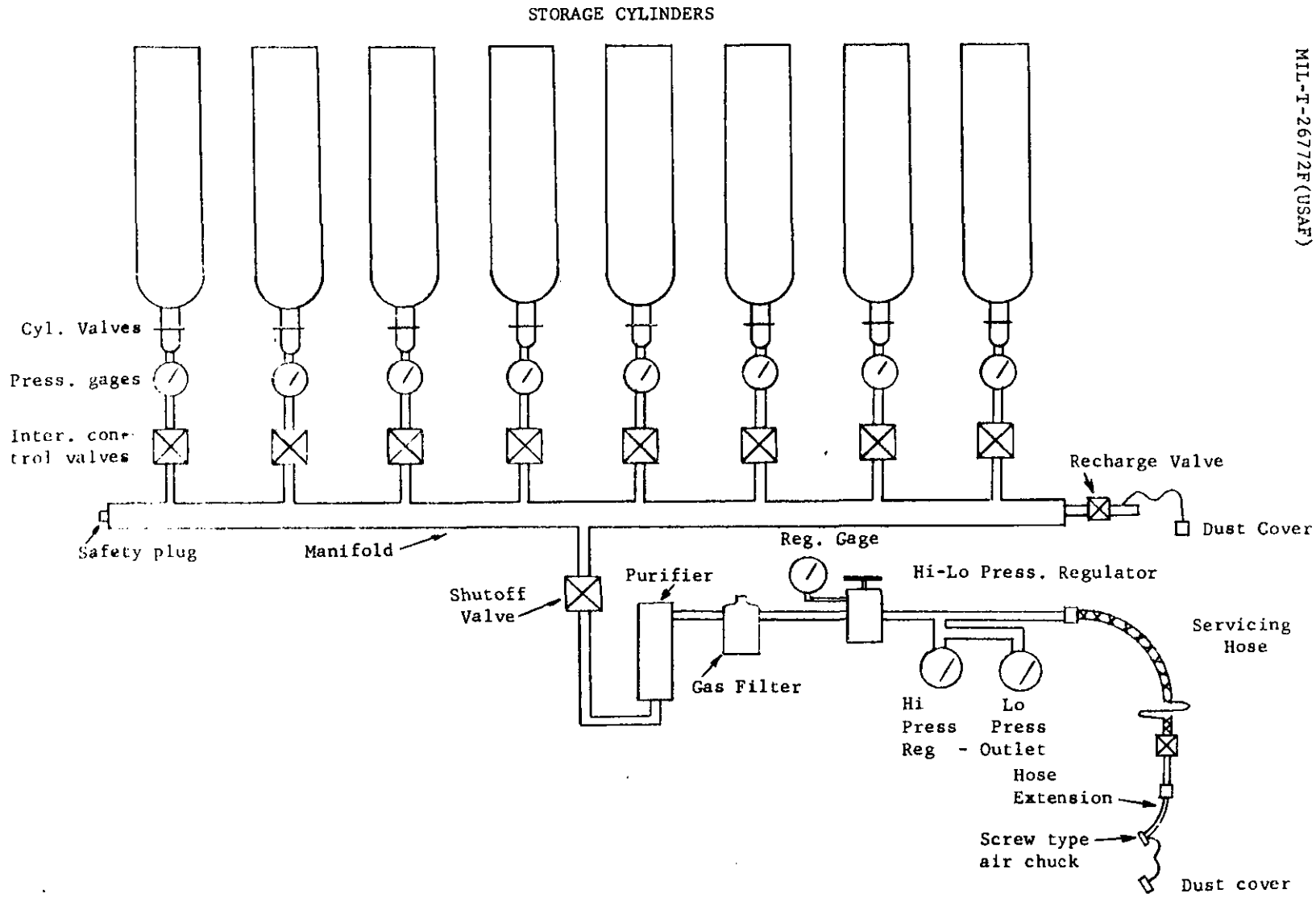


FIGURE 2 - SCHEMATIC FLOW DIAGRAM

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:12Z
Check the source to verify that this is the current version before use.

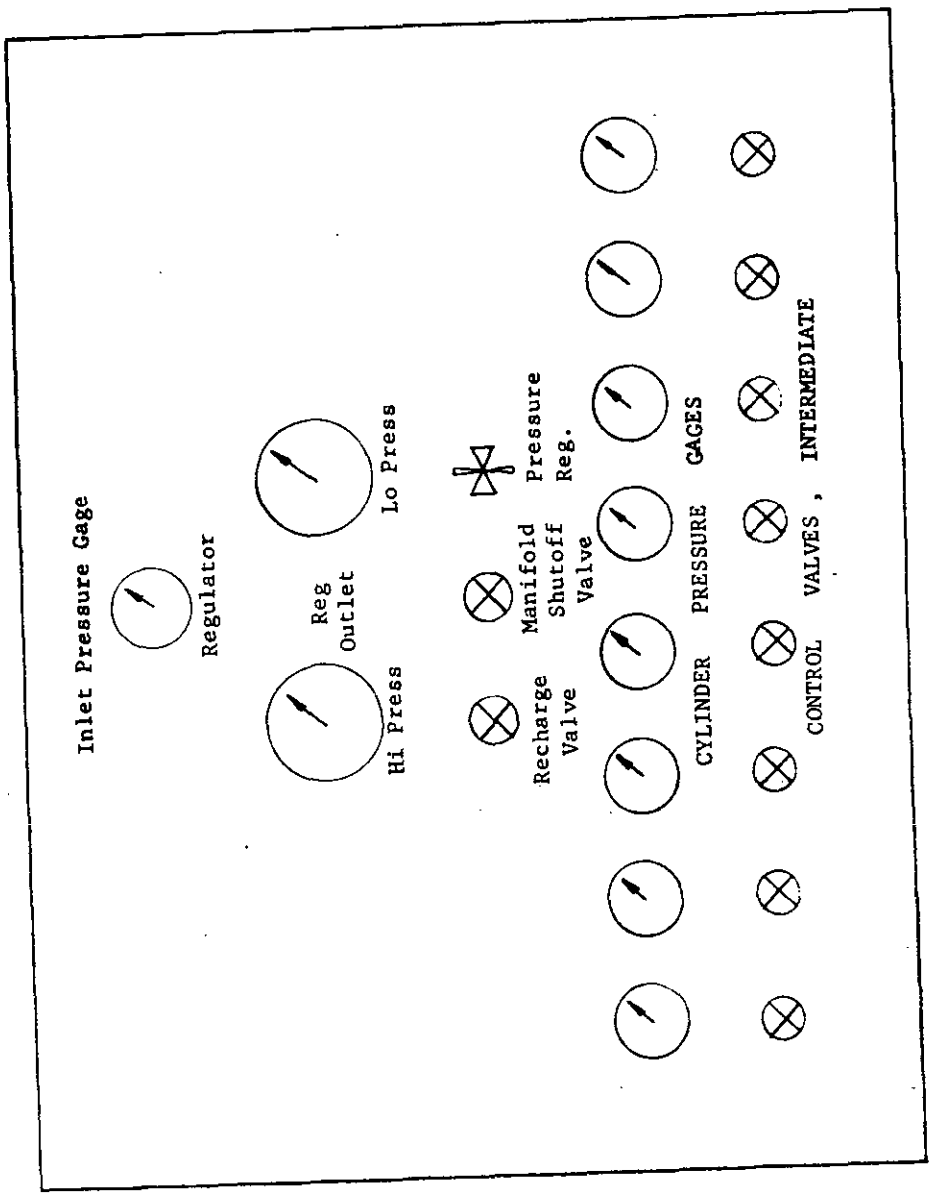


FIGURE 3 - CONTROL AND INSTRUMENT PANEL

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:12Z
Check the source to verify that this is the current version before use.

3.5.6 Common parts. The trailer shall be designed to provide for the maximum practicable interchangeability of hardware and fastening devices by using the minimum number of types and sizes of bolts, screws, nuts, washers, and other common hardware.

3.5.7 Protective covers. All cover plates, gaskets and fittings necessary for protection of contained apparatus during operating, storage and shipment of the trailer shall be provided.

3.6 Construction. The trailer shall be constructed to withstand the strains, shocks, vibrations and other detrimental conditions incident to operation, maintenance, shipping and storage with minimum loss of service time for maintenance, repair, and periodic servicing.

3.6.1 Construction. The trailer shall be constructed to withstand the strains, shocks, vibrations and other detrimental conditions incident to operation, maintenance, shipping and storage with minimum loss of service time for maintenance, repair and periodic servicing.

3.6.2 Component mounting. Trailer components which need not be removed or repositioned for use shall be securely mounted to the trailer chassis or other major components in a manner that will insure against damage or unnecessary movement during operation and transport. All sub-bases and other fixtures needed for safe and secure component mounting shall be provided and installed.

3.6.2.1 Center-of-gravity. The trailer overall center-of-gravity shall be as low as practicable.

3.6.3 Connecting line fittings. All connecting line fittings shall be stainless steel conforming to MS33656.

3.6.4 Controls, instruments and accessories. All automatic and manual controls, instruments, fitting, safety devices and accessories needed for safe and efficient operation of the trailer shall be provided.

3.6.4.1 Instruments. All instruments necessary for indicating the information needed in controlling the trailer servicing and recharging operations shall be securely mounted on the control panel in positions readily visible and accessible to operating and maintenance personnel standing in front of the control panel.

3.6.4.2 Controls. Insofar as practicable, all manually operated controls used in normal operation of the trailer shall be located on the control panel.

3.6.5 Control panel. The control panel shall be of sufficient size for installation of the instruments and controls needed in normal operation of the trailer. It shall be so installed on the trailer that all instruments, controls, et cetera, mounted thereon will be readily visible and accessible to an operator standing in front of the control panel.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:12Z
Check the source to verify that this is the current version before use.

3.6.6 Parts fabrication. Sheet metal parts shall be of such thickness, rigidity and strength as to withstand the dents, warping, vibration, radiated heat, et cetera, encountered in service and maintenance and under the conditions specified herein.

3.6.7 Locking devices. Where practicable, all screws, pins, bolts, et cetera, shall be equipped with locking devices. Safety wire, self-locking nuts, cotter pins, lockwashers, et cetera, will be acceptable. Where practicable, lockwashers shall be secured to bolts or screws. Cotter pins shall be of a type fabricated from corrosion-resistant steel or other corrosion-resistant material.

3.6.8 Pipe thread compound. Tetrafluoroethylene tape conforming to MIL-T-27730 or other pipe thread sealing materials specifically approved by the procuring activity shall be applied to all pipe thread fittings before assembly.

3.6.9 Lifting and tiedown provisions. The trailer shall be provided with lifting and tiedown rings conforming to Drawing 4887796. Installation shall be such as to permit hoisting of the complete trailer, considering vertical accelerations of 3g, or tying it down for safe transport by train, air or water when it is subjected to the acceleration forces specified in MIL-A-8421. The lifting and tiedown rings shall be located at accessible points on the trailer which will permit cable or chain attachment for hoisting and tiedown without damage to any components of the trailer.

3.7 Performance.

3.7.1 Working pressure. The trailer shall have a working pressure (see 6.3.3) of 3500 psi. It shall be capable of servicing high and low pressure aircraft or missile pneumatic systems with air or nitrogen within the range of 25 through 3500 psi inclusive.

3.7.2 Hydrostatic pressure. All parts of the manifold, piping system, and other components, except the gas storage cylinders, safety devices, servicing hose assembly, gages and regulator shall be capable of withstanding a hydrostatic pressure of 5,840 psi for 1 minute without evidence of distortion, damage or leaks.

3.7.3 Total leakage. When the trailer is fully changed to 3500 psi with compressed air or nitrogen at ambient temperature and the servicing system is pressurized to maximum working pressure out to the end of the air chuck specified in 3.8.4.4.3.1, the average leakage rate from the trailer shall not exceed 1 cubic foot when measured over a 24 hour period.

3.7.4 Servicing rate. With the storage cylinders charged at 3500 psi at a temperature of 70° F, the trailer shall be capable of discharging the stored air or nitrogen through the complete servicing system, including the servicing hose, valve, hose extension, and air chuck to a final cylinder pressure of 400 psi in a maximum period of 30 minutes with the regulator preset at a discharge pressure of 3000 psi.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:12Z
Check the source to verify that this is the current version before use.

3.7.5 Recharging rate. The recharging system (see 3.8.3.5) shall provide for filling the trailer at a rate of not less than 50 scfm with a pressure drop across the recharging system not to exceed 250 psi when the cylinders are at atmospheric pressure.

3.7.6 Environmental conditions. The trailer shall be capable of performing as specified herein under the following conditions:

a. Operating temperatures ranging from -65° to +130° F
b. Airborne sand and dust particles
c. Atmosphere containing salt-laden moisture
d. Freezing rain.

3.7.6.1 Storage temperatures. The trailer shall not be damaged by storage in temperatures ranging from -80° to +160° F.

3.7.7 Mobility. The trailer shall be capable of type II, group A mobility in accordance with MIL-M-8090 (see 6.2).

3.7.7.1 Landing wheels. A retractable landing wheel with height adjustment will be supplied in lieu of the Acme-screw-type landing gear specified by MIL-M-8090.

3.7.8 Air transportability. The trailer shall be capable of complying wit the air transportability requirements specified in MIL-A-8421 when filled to design working pressure of 3500 psi with compressed air or nitrogen.

3.7.8.1 The fully loaded trailer shall also be capable of withstanding like forces as imposed by the flight and taxiing loads specified in MIL-A-8421 while being towed or while being transported in a railroad car, semitrailer, et cetera.

3.8 Details of components.

3.8.1 Chassis. The trailer chassis shall be designed and constructed to support and maintain alignment of the high-pressure cylinders and other components specified herein and as shown on figure 1. Flooring shall be provided for the space between the forward bulkhead and the control panel housing as shown on figure 1. A positive means shall be provided to prevent displacement of, or damage to, any components when the trailer is subjected to the conditions specified.

3.8.1.1 Wheels, tires, tubes and hubs. The two main wheels shall be equipped with size 6.00 by 9, 6-ply tires and tubes. The retractable third wheel support shall be of the two position type with a handcrank for elevating to the locked up and down position. The wheel shall be capable of carrying 3 times the imposed load, easily railing over temporary metal runways and of the non-sparking type. Main wheels, hubs, tires and tubes shall conform to MIL-W-8009. Tires will be free running, non-directional, rib type tread tires.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:12Z
Check the source to verify that this is the current version before use.

3.8.1.2 Weight distribution. The weight shall be so distributed that when the fully loaded trailer is parked on a level surface not more than 175 pounds nor less than 165 pounds force, applied at the lunette eye, is required to raise the landing gear clear of the supporting surface.

3.8.1.3 Parking brake. Two wheel parking brakes shall be provided conforming to MIL-B-22182. The parking brakes shall be capable of locking the wheels so that the tires will slide relative to the ground and not roll on dry, level concrete when the trailer is subjected to a towing force sufficient to move the vehicle. The parking brakes shall be capable of holding the loaded trailer stationary on a 30-percent grade.

3.8.2 Storage cylinders. Eight storage cylinders conforming to ICC publication 49 CFR 71-78, Specification 3AA, designed for a working pressure of 3500 psi, shall be installed on the test trailer only unless otherwise specified as shown on figure 1. Each cylinder shall have an internal water volume of not less than 2630 cubic inches. Outside dimensions shall not exceed 51 inches in length and 9 1/2 inches outside diameter. The cylinders shall be designed and constructed in accordance with the general requirements of RR-C-901.

3.8.2.1 Cylinder pressure gages. Individual pressure gages shall be provided for each storage cylinder. The gages shall have a pressure range of from 0 to 5000 psi and shall be not less than 2 1/2 inches in diameter. The gages shall be a part of the manifold system (see 3.8.3) and shall be connected between the cylinders and the intermediate control valves as shown on figure 2. Each gage shall have a minimum accuracy of $\pm$ 2 percent of the scale range. Each gage face shall have figure intervals indicating each 500 psi increment through out the entire range. In addition, each gage face shall have inyermediate intervals indicating 100 psi increments throughout the entire range.

3.8.2.1.1 Cylinder valves. Cylinders will be furnished compatible cylinder valves.

3.8.2.2 Cylinder mounting racks. The cylinders shall be mounted lengthwise on the trailer so that each cylinder can be removed separately without disturbing the other cylinders. This shall be accomplished by mounting the cylinders on individual troughs with adequate supporting framework, one cylinder above the other as shown on figure 1. The troughs shall be adequately perforated to prevent accumulation of water therein in the normal position of the trailer. Provisions shall be made to permit easy loading of the cylinders with rollers or other devices.

3.8.2.2.1 Clamps. A quick acting and easily accessible bar arrangement, designed to hold the cylinders in place shall be provided.

3.8.2.3 Bulkhead. A steel bulkhead shall be provided on the trailer chassis at front end of the cylinder assembly. The bulkhead shall contain contoured holes through which the cylinder necks protrude.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:12Z
Check the source to verify that this is the current version before use.

3.8.2.3.1 The bulkhead shall be of sufficient overall width and height to extend not less than 2 inches beyond the extreme cylinder outline as viewed from the end. Bulkhead thickness shall be sufficient to retain the cylinders in the proper longitudinal position and insure against movement when the trailer is subjected to the conditions specified herein.

3.8.2.3.2 Bulkhead bottoms shall be secured to the chassis and the tops braced by compression or tension members so that longitudinal movement of the cylinders will not occur under the conditions specified herein. Bulkheads shall be installed to be readily demountable without the use of special tools and without breaking welds or seals.

3.8.3 Manifold system. The cylinder manifold system shall be designed for a working pressure of 3500 psi. The cylinders shall be manifolded together and connected to the servicing system as shown on figure 2.

3.8.3.1 Manifold connecting lines. Stainless steel connecting lines of not less than 1/4 inch inside diameter shall be provided between the storage cylinders and the manifold, and between the manifold and the servicing hose connection as shown on figure 2. Each connecting line shall be fabricated and installed to insure against applying damaging stresses to the manifold piping or the line connections as a result of the relative movement between the trailer components that occurs during transport and use in servicing operations.

3.8.3.2 Intermediate control valves. Intermediate control valves shall be Circle Seal Part Number B933T-2PP(P) or equal.

3.8.3.3 Manifold and storage cylinder safety devices. The manifold and each compressed gas storage cylinder shall be protected against excessive pressuree with metal rupture discs conforming to CGA Pamphlet S-1 as shown on figure 2. The rupture disc shall be type CG-1 of the unbacked, frangible type. The disc bursting pressure shall be 5,000 $\pm$ 250 psi.

3.8.3.4 Manifold shutoff valves. A manually operated shutoff valve shall be Circle Seal Part Number B933T-2PP(P) or equal.

3.8.3.5 System recharging valve assembly. A system recharging valve assembly shall be provided in the manifold piping as shown on figure 2. The system recharging valve assembly shall consist of a recharging valve and a recharging disconnect fitting. The recharging valve outlet shall be directly connected to the manifold. The recharging disconnect fitting shall be connected to the recharging valve inlet.

3.8.3.5.1 Recharging valve. The recharging valve shall be a manually operated shutoff valve designed and constructed in accordance with 3.8.3.2.

3.8.3.5.2 Recharging disconnect fitting. The recharging disconnect fitting shall be designed, constructed and installed for ready and leaktight attachment of the hose specified in paragraph 3.8.4.4.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:12Z
Check the source to verify that this is the current version before use.

3.8.3.5.2.1 Disconnect fitting mounting. The recharging disconnect fitting shall be so mounted that force applied in attaching or removing a recharging hose or while a recharging hose is attached will not damage the manifold system piping.

3.8.3.5.2.2 Protective cap. The recharging disconnect fitting shall be provided with a protective cap that will tightly close and protect the fitting when a recharging hose is not attached. The protective cap shall be attached to the disconnect fitting by a chain or other device to protect against accidental loss.

3.8.4 Servicing system. The servicing system shall provide for servicing the compressed gas from the high-pressure storage cylinders and manifold into the system being charged. The servicing system shall consist of a purifier, filter, pressure regulator, and a servicing hose assembly.

3.8.4.1 Purifier cylinder. A purifier cylinder conforming to MIL-C-4810 shall be provided in the servicing system as shown on figure 2. Installation of the purifier cylinder shall provide for ready access through the panels specified in 3.8.7.1 for cartridge change and maintenance.

3.8.4.2 Gas filter assembly. A gas filter assembly designed to use, and equipped with, an AN6235-3A or AN6235-4A filter element shall be provided in the servicing system as shown on figure 2. The filter assembly shall have a working pressure of 3500 psi. Design and construction of the filter assembly shall insure against gas bypassing the filter element, and provide for easy and convenient replacement of the filter element.

3.8.4.3 High-low pressure regulator. The pressure regulator shall be Victor Model BLR 10 or equal and shall meet the test requirements of Section 4.

3.8.4.3.1 Overpressure relief device. A differential-type, overpressure relief device shall be provided on the low-pressure side of the regulator. The relief device shall start to vent before the downstream pressure reaches 100 psi above any set dead-end pressure up to 3500 psig and flow shall reach at least 30 scfm before the downstream pressure reaches 400 psi above the dead-end pressure. The relief device shall reseal after venting when the outlet pressure is reduced to not less than the set dead-end delivery pressure.

3.8.4.3.2 Regulator pressure gages. The pressure regulator installation shall be equipped with a regulator gauge and high and low outlet pressure gages.

3.8.4.3.2.1 Regulator gage. The regulator gage shall be identical to the cylinder pressure gages.

3.8.4.3.2.2 High-pressure outlet gage. A pressure gage with a dial diameter not less than 4-1/2 inches shall be provided for indicating the regulator outlet pressure as shown on figure 1. The gage shall have a scale range of 0 to 5,000 psi, shall be accurate to within one-half of 1 percent of full scale, and shall have a safety back. The gage face shall have figure intervals, indicating each 500 psi increment plus intermediate interval graduations indicating 50-psi increments throughout the entire range.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:12Z
Check the source to verify that this is the current version before use.

3.8.4.3.2.3 Low-pressure outlet gage. A pressure gage with a dial diameter not less than 4 1/2 inches shall be provided for indicating the regulator low outlet pressures as shown on figure 1. The gage shall have a scale range of 0 to 600 psi, shall be accurate to one-half of 1 percent of full scale, and shall have figure intervals indicating each 50 psi increment plus intermediate interval graduations indicating 5 psi increments throughout the entire range. Provisions shall be made to insure against damage to the gage when the servicing line is pressurized to 3500 psi.

3.8.4.3.3 Filter screen. A filter screen which will retain all particles of 200 mesh or larger size shall be provided in the regulator inlet to protect the mechanism.

3.8.4.4 Servicing hose assembly. A servicing hose assembly consisting of a servicing hose, a servicing valve, and a hose extension shall be provided in the servicing system as shown on figure 2.

3.8.4.4.1 Servicing hose. The servicing hose shall be a 30 ± 1 foot long, size 4, pneumatic hose assembly conforming to MIL-H-26666.

3.8.4.4.2 Servicing valve. A servicing valve conforming to MIL-V-4719, except as specified below, shall be installed on the servicing end of the servicing hose as shown on figure 2 to control the flow of gas during servicing operations. The MIL-V-4719 requirement changes applicable to the servicing valve are:

a. The servicing valve shall be designed for use with air or nitrogen.

b. Where applicable, markings denoting air or nitrogen shall be used in lieu of oxygen markings.

c. Wherever specified, the working pressure shall be 4000 psi in lieu of 2200 psi.

d. Wherever specified, the hydrostatic test pressure shall be 6670 psi in lieu of 3500 psi.

e. Wherever specified, the number of opening and closing cycles required for endurance shall be 1500 in lieu of 500.

3.8.4.4.3 Hose extension. The hose extension shall be a 9 +2, -0 inch long, -4 size pneumatic hose assembly conforming to MIL-H-26666. The trailer end of the hose extension shall be installed on the servicing side of the servicing valve as shown on figure 2. The air chuck shall be installed on the servicing end of the hose extension.

3.8.4.4.3.1 Air chuck. The screw-on-type air chuck shall conform to Drawing 44A19817 and details thereto for attachment to the -4 size hose extension specified in 3.8.4.4.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:12Z
Check the source to verify that this is the current version before use.

3.8.4.4.3.1.1 Protective cap. The air chuck shall be provided with a protective plug or cap that will tightly close and protect the fitting when it is not in use for servicing. The protective plug or cap shall be attached to the air chuck by a chain or other device to protect against accidental loss.

3.9 Dimensions. The overall outside dimensions of the trailer shall not exceed the limits shown on figure 1.

3.10 Weight. The gross weight of the complete trailer, including high-pressure gas storage cylinders, shall not exceed 2500 pounds.

3.11 Finishes and protective coatings.

3.11.1 Exposed parts and surfaces. Exposed parts and surfaces of the trailer shall be cleaned, treated and finished as specified in MIL-STD-808. When more than one finish process is permitted in MIL-STD-808 for a part or surface, the most applicable process shall be determined by the contractor unless otherwise specified herein.

3.11.2 Finish. Final film color designation DG shall apply to all trailer parts and surfaces.

3.12 Operational markings.

3.12.1 Operating and precautionary instruction plate. A permanently and legibly marked operating and precautionary instruction plate shall be provided in a conspicuous location on or near the control panel. Clear, concise, step-by-step instructions detailing all necessary operating procedures shall be included. Instruction plates shall be made from sheet aluminum or sheet zinc of not less than 0.050 inch thickness, anodized or etched to produce raised markings, with a black or other dark color background, and with a raised border of not less than 1/8 inch width.

3.12.2 Identification plates. All manually controlled items and gages located on the forward side of the control panel shall be identified with individual plates. The plates shall be permanently mounted on the control panel, completely visible to operating personnel, and fabricated in accordance with 3.12.1.

3.12.3 The following information shall be stenciled on the control panel in block-type letters a minimum of 1/2 inch high. Stenciling shall be in accordance with MIL-STD-808, film designation UL (black).

AIR OR NITROGEN GAS ONLY
CAUTION - HIGH PRESSURE.

3.12.4 The inlet and outlet of the high-low pressure regulator shall be permanently marked and an arrow shall be marked on the regulator body to indicate the direction of the flow.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:12Z
Check the source to verify that this is the current version before use.

3.12.5 Transportation data plate. A transportation data plate, as specified in MIL-M-8090, shall be provided.

3.12.6 Registration number. The lettering U.S. AIR FORCE and the registration number furnished by the procuring activity (see 6.2) shall be applied in one line with 1 3/4 inch silver reflective letters on a 2 3/4 inch red reflective background centered on a panel attached to each side of the chassis and across the rear of the trailer frame. Marking shall be accomplished by applying reverse silk screened reflectorized tape conforming to MIL-R-13689.

3.12.7 Lubrication points. Each lubrication point shall be identified by a filled-in circle, approximately 2 inches in diameter, around the lubrication point. The circle shall be painted on the equipment with gloss enamel conforming to MIL-STD-808, insignia red, film designation BG.

3.12.7.1 Lubrication warning plates. Where the use of high-pressure lubricating equipment will cause damage to grease seals or other parts, a warning plate shall be affixed to the equipment in a conspicuous location.

3.12.8 Air pressure. The proper air pressure to be maintained in the trailer tires shall be stenciled in block letters approximately 1/2 inch high near the wheels.

3.13 Identification of product. Equipment, assemblies and parts shall be marked for identification in accordance with MIL-STD-130.

3.13.1 Trailer nameplate location. The trailer nameplate shall be securely attached to the outside of the control housing assembly in a readily visible location at the front of the trailer.

3.14 Workmanship. The trailer, including all parts and accessories, shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to the following:

a. Freedom from blemishes, defects, burrs, and sharp edges

b. Accuracy of dimensions, radii of fillets, and marking of parts and assemblies

c. Thoroughness of soldering, welding, brazing, painting, wiring and riveting

d. Alignment of parts and tightness of assembly screws and bolts, et cetera.

3.14.1 Riveting. Riveting operations shall be performed to insure that the rivets are tight and satisfactorily headed.

3.14.2 Cleaning. The trailer shall be thoroughly cleaned. Loose, spattered or excess solder; metal chips; and other foreign material shall be removed during and after final assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:12Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the trailer shall be classified as follows:

a. Preproduction testing (see 4.4)
b. Acceptance tests (see 4.5).

4.3 Test conditions. Unless otherwise specified, tests shall be conducted under ambient conditions (see 6.3.4).

4.3.1 Instrumentation.

4.3.1.1 Pressures. Pressures and pressure differentials shall be measured by means of Bourdon-type gages. Wherever practicable, the gages furnished as part of the trailer shall be utilized.

4.3.1.1.1 Pressures shall be recorded in pounds per square inch gage.

4.3.1.2 Temperatures. Temperatures shall be measured by appropriately located thermometers or thermocouples used with calibrated potentiometers.

4.3.1.2.1 Temperatures shall be recorded in degrees Fahrenheit.

4.3.1.3 Barometric pressure. Barometric pressure shall be measured by a properly calibrated mercurial or aneroid barometer.

4.3.1.4 Gas flow rates. Gas flow rates shall be measured with standardized nozzles or orifices as specified in ASME Power Test Code PTC 19.5:4.

4.3.1.4.1 Gas flow rates shall be recorded as standard (29.92 inches Hg and 70° F) cubic feet per minute.

4.3.2 Test data.

4.3.2.1 Correction of data. If the tests specified herein cannot be conducted under the stated conditions, the tests may, upon approval by the procuring activity, be conducted under other conditions and the performance under the specified conditions calculated from the test results obtained. The manner of calculation methods used shall be presented for review by the procuring activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:12Z
Check the source to verify that this is the current version before use.

4.3.2.2 Operational data. Wet and dry bulb temperature readings shall be recorded when specified for an individual test, and shall be taken at the intervals specified therefor.

4.3.2.3 Psychrometric data. Wet and dry bulb temperature readings shall be recorded when specified for an individual test, and shall be taken at the intervals specified therefor.

4.3.2.4 Barometric pressure. The barometric pressure shall be measured and recorded when specified for an individual test, and shall be obtained at the intervals specified therefor.

4.3.2.5 Tolerances.

4.3.2.5.1 Data on gage pressures shall be accurate to within 1 percent.

4.3.2.5.2 Data on temperatures shall be accurate to within 2° F.

4.3.2.5.3 Data on barometric pressure shall be accurate to within 0.01 inch Hg.

4.3.2.5.4 Data on gas flow rates shall be accurate to within 1 percent.

4.3.3 Inspection, servicing and adjustments. Inspections, servicing and adjustments shall be performed as specified by the manufacturer unless such instructions are contrary to those necessary for compliance with the requirements specified herein.

4.3.4 Observations. During the progress of all tests, the performance of the trailer shall be observed to determine whether it complies with all requirements specified herein.

4.3.5 Preparation for test. The trailer shall be prepared for test by assembly, adjustment and servicing as specified in the instruction manual.

4.3.6 Pressure test record. A record of the pressures at which each trailer component is tested and the length of time during which it is subjected to the pressure shall be retained for examination by the procuring activity.

4.4 Preproduction testing.

4.4.1 Preproduction test sample. One trailer shall be subjected to the tests specified in 4.4.3.

4.4.2 Reliability and maintainability information. The following information shall be submitted as an attachment accompanying the preproduction test report (same quality and type of copies) or shall be included as part of that report:

a. A complete summary of the data resulting from the maintainability verification studies required by 4.6.23.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:12Z
Check the source to verify that this is the current version before use.

b. All failures, maintenance, servicing, irregular operations and other similar events of the testing shall be identified by a brief description of the action and by the accumulated operating hours, cycles, or position in the test procedures, as appropriate. Test conditions during the failures or irregular operations identified shall be included.

c. A description of the engineering reasoning behind, and of any tests conducted to determine assignable causes for, all failures and irregular operations identified.

d. A description of the engineering reasoning behind any corrections made, to be made on production items, or proposed to be made, and the predicted effectiveness of these corrections to improve the reliability, operability, maintainability, and usefulness of the item.

e. Test activity or contractor comments on features that make unduly difficult or repetitive any maintenance, servicing, or adjustments that were performed or that are anticipated in field use.

f. Test activity or contractor comments on features that, if modified, should improve the usefulness, reliability, or maintainability of the item.

g. Test activity or contractor comments on field conditions or procedures to be avoided or cultivated to increase the reliability, maintainability and useful life of the item.

h. The manhours (± 5 percent of actual experienced) required for each maintenance or servicing action performed during the tests. A brief description of the qualifications and experience of the personnel involved shall be included and shall be adequate to allow comparison to the personnel anticipated to be assigned for similar fieldwork.

i. Estimates of the manhours anticipated for other servicing and maintenance actions anticipated in field use. A brief description of the qualifications and skills required to accomplish these servicing and maintenance actions shall be included.

j. References to other documents that have supplied or will supply any of the above information to the procuring activity.

4.4.3 Preproduction tests. The preproduction tests shall consist of all the tests described under 4.6.

4.5 Acceptance test. Acceptance tests shall consist of the individual tests.

4.5.1 Individual tests. Each trailer shall be subjected to the following tests as described under 4.6:

a. Examination of product (see 4.6.1)
b. Mechanical inspection (see 4.6.2)
c. Operation check (see 4.6.3)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:12Z
Check the source to verify that this is the current version before use.

d. Hydrostatic tests (see 4.6.4)
e. Total leakage test (see 4.6.6)
f. Cleaning effectiveness tests (see 4.6.23).

4.6 Test methods.

4.6.1 Examination of product. The overall trailer and its individual components shall be inspected to determine compliance with the requirements specified herein. This inspection shall include examination of the general design, arrangement of accessories and major assemblies, quality of materials and workmanship, finish and appearance of all parts and the ease of adjustment and alignment of all parts.

4.6.2 Mechanical inspection. A mechanical inspection of all components and parts of the trailer shall be conducted. All pertinent data concerning conditions, defects of manufacture, damage in transit and damage through use prior to tests shall be recorded. This inspection shall include servicing of the trailer for operational use.

4.6.3 Operation check. The servicing system shall be operated and thoroughly checked to insure that the manifold system, gages, valves and regulator are functional and that all components operate as intended.

4.6.4 Hydrostatic test. All parts of the trailer that operate at pressure, except the cylinders, safety devices, regulator, gages and servicing hose assembly shall be subjected to an internal pressure of not less than 5,840 psi for not less than 1 minute. Distortion, damage, or leaks resulting from the pressure testing shall be cause for rejection of the components affected. Following the pressure test, all components shall be thoroughly cleaned and dried to insure complete removal of the pressure test fluid.

4.6.5 Dimensions and weight. The overall dimensions and total weight, including all equipment and accessories, of the completely assembled trailer shall be recorded on the test data sheets. The dimensions and weight shall not exceed the limits specified in 3.10 and 3.11.

4.6.6 Total leakage test. A calibrated laboratory precision type pressure gage having a demonstrated accuracy of one-half of 1 percent and capable of being read to $\pm$ psi shall be connected to the trailer manifold for this test. The trailer shall be charged to 3,500 psi and the chuck at the end of the servicing assembly shall be plugged. The servicing hose assembly shall be pressurized to 3,000 psi with the controls provided on the trailer. With all valves open except the recharging valve, the cylinders charged to 3500 psi, the servicing assembly pressurized, and no further adjustment of valves or regulator, the trailer shall remain at a uniform ambient temperature $\pm 5^{\circ}$F for 24 hours. A pressure loss in excess of 39 psi or leakage in excess of 24 standard cubic feet of gas in a 24 hour period shall be cause for rejection.

4.7.7 Recharging rate test. The cylinder-to-manifold leads shall be disconnected from the cylinders for this test. Air or nitrogen shall be supplied

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:12Z
Check the source to verify that this is the current version before use.

to the recharging connection specified in 3.8.3.5.2 through a hose of the type specified in 3.8.4.4.1 at not less than 50 scfm and discharged to atmosphere through the manifold cylinder leads. A pressure drop in excess of 250 psi between the chuck and the cylinder connections shall be cause for rejection.

4.6.8 Servicing rate test. The storage cylinders shall be charged to a pressure of not less than 3500 psi at $70^{\circ} \pm 10^{\circ}$ F. The regulator shall be set for a discharge pressure of 3000 psi. Without disturbing the regulator setting, the gas shall be permitted to flow through the complete servicing system, including the servicing hose, servicing valve, extension and chuck, with the cylinder shutoff, manifold shutoff and servicing valves fully opened, until the cylinder pressure drops to 400 psi. The total time elapsed from beginning of flow until the cylinder pressure drops to 400 psi shall be recorded, and shall not exceed 30 minutes or the trailer shall be considered to have failed the test.

4.6.9 Manifold shutoff and recharging valve tests. Three valves of each type shall be subjected to the following tests in the order specified:

4.6.9.1 Pressure test. With the valve open and the outlet capped, 3500 psi gas pressure shall be applied at the inlet and the valve submerged in water for 10 minutes. There shall be no evidence of leakage. This test shall be repeated with the valve closed to not more than 30 pound-inches torque and with the outlet open. Leakage exceeding 2.5 standard cubic inches per hour shall be cause for rejection.

4.6.9.2 Overtorque. The valve shall be closed to not less than 150 pound-inches torque for 10 minutes, then opened and reclosed to not more than 30 pound-inches torque. The valve shall then be tested in accordance with the valve closed portion of 4.6.9.1. Total leakage exceeding 2.5 standard cubic inches per hour shall be cause for rejection.

4.6.9.3 Endurance. With the valve outlet capped, 3500 psi gas pressure shall be applied and maintained at the inlet. The valve shall then be subjected to not less than 2500 opening and 30 pound-inches torque closing cycles without interim valve lubrication, adjustment, or repair (see 6.3.5). The test shall be repeated for 2500 cycles with the valve outlet open to atmosphere. At the conclusion of cycling tests, and without any adjustment and with the outlet open, 3500 psi gas pressure shall be applied at the inlet. Total leakage exceeding 2.5 standard cubic inches per hour shall be cause for rejection.

4.6.10 Servicing valve test. The servicing valve shall be tested in accordance with, and fulfill, the requirements of MIL-V-4719 except that:

a. Wherever specified, the working pressure shall be 4000 psi in lieu of 2200 psi.

b. Wherever specified, hydrostatic tests shall be 6670 psi in lieu of 3500 psi.

c. Wherever specified, the number of opening and closing cycles required for endurance testing shall be 1500 in lieu of 500.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:12Z
Check the source to verify that this is the current version before use.

4.6.11 Intermediate control valve tests. Three of the intermediate control valves shall be subjected to the following tests in the order specified:

4.6.11.1 Flow tests. With the valve outlet open to the atmosphere, a gas pressure of not more than 100 psi shall be applied to the valve inlet. A flow rate of less than 25 scfm through the valve under the conditions specified shall be cause for rejection. The actual flow rates determined for each valve tested shall be recorded and reported.

4.6.11.2 Endurance. With the valve outlet open to the atmosphere, 3500 psi gas pressure shall be applied to the valve inlet, and the valve shall then be subjected to not less than 1500 complete opening and closing cycles without interim lubrication, repair, or adjustment. Force required to operate the intermediate control valve in excess of easy manual, one ungloved hand operation without tools at the conclusion of the cycling tests shall be cause for rejection.

4.6.11.3 Pressure test. With the valve closed and the outlet open, 3500 psi gas pressure shall be applied and maintained at the inlet and the valve submerged in water for 10 minutes. Leakage exceeding 2.5 standard cubic inches per hour shall be cause for rejection.

4.6.12 High-low pressure regulator test. In addition to the operational tests required for the complete trailer, the high-low pressure regulator shall be tested for compliance with 3.8.4.3 as specified in 4.6.12.1 through 4.6.12.5. Laboratory precision instruments shall be used for the tests. All pertinent data shall be recorded.

4.6.12.1 Hydrostatic pressure. The high and low pressure zones of the regulator, exclusive of pressure gages and safeyy relief devices, shall be subjected to a hydrostatic pressure of 5840 psig. After hydrostatic testing, the regulator shall be disassembled and inspected. Signs of leakage during the test or indications of permanent deformation of internal or external components shall be cause for rejection.

4.6.12.2 Pressure range. With an inlet pressure of 3600 psig, the regulator shall be set for 0 psig dead-end delivery pressure. With the outlet closed, the outlet pressure shall be increased to 3500 psig in steps of 25 psi, and then decreased to 0 psig in steps of 25 psi using the control knob. The test shall be repeated not less than four times. A torque in excess of 50 pound-inches required on the control knob to change the outlet pressure from 0 to 3500 to 0 psig shall be cause for rejection.

4.6.12.3 Low-flow outlet pressure variation. With an inlet pressure of 3500 psig to the regulator, the delivery pressure shall be adjusted to 500 psig at a flow rate of 10 scfm. Without further adjustment of the regulator, the inlet pressure shall be slowly reduced to 800 psig. The test shall be repeated with the temperature of the regulator stabilized at -65° F, and again at +130° F. During testing, outlet pressure variation in excess of 25 psi for any 500 psi increment reduction in let pressure shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:12Z
Check the source to verify that this is the current version before use.

4.6.12.4 High flow outlet pressure variation. The regulator shall be subjected to an inlet pressure of 3000 psig and the dead-end delivery pressure adjusted to 2000 psig. With a constant 3000 psig inlet pressure and no further adjustment of the regulator, a downstream shutoff valve shall be opened until a flow of 75 scfm through the regulator is obtained. The delivery pressure shall not drop to less than 1900 psig at 75 scfm flow and shall not exceed 2000 psig when the flow is again reduced to zero. This test shall be repeated with a constant inlet pressure of 1500 psig and with an original dead-end pressure of 500 psig. The delivery pressure shall not drop to less than 400 psig at 75 scfm flow, and shall not exceed 500 psig when the flow is again reduced to zero. Exceeding the stated limits shall be cause for rejection.

4.6.12.5 Overpressure relief device. With a suitable inlet pressure, the regulator shall be adjusted for a 500 psig, dead-end delivery pressure. Without further adjustment of the regulator, a source of higher pressure gas shall be connected to the regulator outlet and permitted to flow into the low-pressure zone to the regulator. The pressure at which the overpressure relief device starts to vent shall be recorded, and the flow rate of the effluent gas shall be measured as the outlet pressure increases. The source of higher pressure gas shall be removed from the regulator outlet and the pressure at which the overpressure relief device reseals shall be noted. An initial start-to-vent pressure of more than 600 psig, an effluent gas flow rate of less than 30 scfm at 900 psig, or a resealing pressure of less than 500 psig shall be cause for rejection.

4.6.13 Outlet pressure gage tests.

4.6.13.1 Pressure cycling. The servicing piping between the regulator and the servicing hose inlet, including both high and low pressure servicing gages, shall be pressurized to not less than 3500 psi for not less than 1 minute. The test pressure shall then be reduced to atmospheric. The test cycle shall be repeated 10 times.

4.6.13.2 High pressure outlet gage tests. Following the pressure cycling specified in 4.6.13.1, the high pressure gage shall be checked on a dead weight tester at 100 psi intervals from 0 through 3500 psi for compliance with the accuracy requirements of 3.8.4.3.2.2. An error in excess of that specified at any pressure shall be cause for rejection. All gage data shall be recorded and reported.

4.6.13.3 Low pressure outlet gage tests. Following the pressure cycling specified in 4.6.13.1, the low pressure outlet gage shall be checked on a dead weight tester at 50 psi intervals from 0 through 600 psi for compliance with the accuracy requirements of 3.8.4.3.2.3. An error in excess of that specified at any pressure shall be cause for rejection. All gage test data shall be recorded and reported.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:12Z
Check the source to verify that this is the current version before use.

4.6.14 Safety device burst tests. Not less than five samples of each type frangible disc furnished to comply with 3.8.3.3 shall be tested as specified in CGA Pamphlet S-1 to determine their burst pressure. The pressure at which each test disc bursts shall be reported to the nearest 50 psi. Failure of any sample to burst within the pressure specified shall require application of the same test to 10 additional samples. Failure of 2 out of the 15 samples tested shall be cause for rejection of the entire lot.

4.6.15 Weight distribution test. The fully loaded trailer with landing wheel extended shall be parked on a level surface. Force shall be applied at the lunette eye to raise the landing wheel clear of the supporting surface. A force required of more than 75 pounds or less than 65 pounds shall be cause for rejection. The actual force required shall be recorded and reported.

4.6.16 Mobility tests. The trailer shall be subjected to the general tests of MIL-M-8090 as specified for all types of mobility and to the tests specified therein for type II, group A vehicles.

4.6.17 Air transportability test. A complete trailer assembly, including the high pressure gas storage cylinders, shall be tested for air transportability as specified in MIL-A-8421.

4.6.18 Air servicing to 3,000 psi reliability demonstration. A compressed gas cylinder suitable for 3000 psi having an internal water volume of not less than 60 cubic inches shall be used for this test. The cylinder shall be provided with a charging connection of the type specified in 3.8.3.5.1 to accept the hose specified in 3.8.4.4.1. This cylinder hereinafter shall be called the test cylinder. The trailer shall be charged to 3300 $\pm$ 200 psi. All trailer valves shall be closed and the recharging valve opened to relieve any pressure upstream of the manifold shutoff valve and the recharging valve reclosed. Connect the hose at the end of the servicing hose to the test cylinder in preparation for charging it to not less than 3000 psi as follows:

a. Fully open a cylinder valve. The servicing pressure regulator shall be set for a no-flow, no-pressure condition against the servicing valve.

b. Fully open the manifold shutoff valve.

c. Adjust the regulator for a deadend pressure of not less than 3000 psi.

d. Slowly open the servicing valve to the fully open position to charge the test cylinder to not less than 3000 psi. The servicing valve shall remain open for not less than 5 minutes after a 3000 psi charge is obtained. An increase of pressure in excess of 25 psi over the deadend set pressure in item (c), or venting of the regulator overpressure relief valve shall be considered as failure to pass the test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:12Z
Check the source to verify that this is the current version before use.

e. By use of the regulator control, bleed the test cylinder completely.

f. Close the servicing valve.

g. Close the cylinder valve.

h. Open the recharging valve to bleed pressure between the regulator and cylinder valve, reclose when regulator inlet pressure has dropped to 0 psi.

i. Close the manifold shutoff valve. A test cylinder charging cycle depleting the trailer cylinders to less than 3100 ± 50 psi shall not be considered a valid cycle and will require use of another cylinder or discontinuance of cycling until the trailer cylinders are recharged to 3300 ± 200 psi. Cycling shall continue until 50 valid test cylinder charging cycles have been successfully completed. Any failure, malfunction or inability of the trailer to complete this air servicing to 3000 psi demonstration shall be cause for rejection.

4.6.19 Servicing and maintenance tests. All normal preventive maintenance and servicing operations specified in the handbook shall be performed to demonstrate their adequacy, ease of accomplishment, and the accessibility of parts and assemblies for performance of same. Insofar as practicable, these tests shall be conducted as part of the normal preventive maintenance, servicing, and inspections performed in accomplishing the testing specified herein. Interferences or obstructions to servicing or preventive maintenance shall be recorded in detail on the test data sheets.

4.6.20 Mechanical check. Upon completion of the above tests, a critical inspection shall be made of components to determine their operability and any damage or undue wear incurred during the tests. Teardown and parts measurement shall be made only in those cases where service life is in question. Where teardown and parts measurement are performed, wear or distortion that exceeds the limits allowed by the manufacturer for new parts shall be cause for considering the part or parts affected as having failed to complete the test satisfactorily.

4.6.20.1 Ferrous parts subject to high stress in operation and that are suspected of having defects shall also be subjected to magnetic particle inspection (magnaflux) and shall exhibit no indications of damage attributable to the tests, or shall be considered as having failed the test. Magnetic particle inspection shall be performed in accordance with MIL-I-6868.

4.6.20.2 Nonmagnetic parts suspected of defects shall be subjected to inspection with fluorescent penetrant (black light) and shall exhibit no indications of defects or damage attributable to the tests that would impair their life or usefulness, or shall be considered as having failed the test. Penetrant inspection shall be performed in accordance with MIL-I-6866.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:12Z
Check the source to verify that this is the current version before use.

4.6.21 Reliability demonstration and test. Satisfactory completion of all tests specified herein demonstrates compliance with the quantitative reliability requirements of this specification.

4.6.22 Maintainability verification. Verification of compliance with the maintainability requirements specified in 3.5.3 shall be accomplished by performing and reporting on the studies detailed in the paragraph entitled Maintenance Task Simulation of appendix B to MIL-M-26512.

4.6.23 Cleaning effectiveness tests.

4.6.23.1 Visual inspection. All equipments, pipes and components that will contact the product carried shall be visually examined for evidence of corrosion products, metal chips, scale, weld scale, oil, grease, paints, preservatives, decals, or other contamination or foreign matter. Any evidence of contamination or foreign matter shall require recleaning and retest.

4.6.23.2 Ultraviolet light inspection. All accessible surfaces and parts, including insofar as possible the interior surfaces of the gas storage cylinders, that will contact the product carried shall be visually inspected for hydrocarbons with ultraviolet light. Any evidence of fluorescence shall require recleaning and retest.

4.6.23.3 Particulate matter test. Following the cleaning and degreasing operations specified herein and equipment final assembly, the trailer shall be filled to not less than 2000 psi with clean, dry air or nitrogen, and all valves closed. The air or nitrogen shall contain not more than 25 parts per million water vapor by volume and not more than 1 ppm by volume of total hydrocarbons expressed as methane equivalent. The trailer front end shall be lowered until the storage cylinder longitudinal axes are inclined at an angle of 45 $\pm$ 5° with the horizontal. The contained air or nitrogen shall then be discharged through the filter specified in 4.6.24.4 in accordance with the following procedure.

a. Connect the filter specified in 4.6.24.4 to the manifold valve outlet so that all gas discharged through the connection must pass through the filter.

b. Discharge gas at maximum flow rate through the manifold outlet valve and filter for a period of not less than 1 minute from each of the trailer gas storage cylinder banks in turn. The flow from one cylinder shall be stopped before beginning discharge from the next.

4.6.23.1 Following the specified gas discharge procedures, the filter shall be removed and thoroughly cleaned. The removed material shall be collected, dried, weighed and examined for size. The presence of any solid particle with any dimension greater than 1000 microns, or more than 25 milligrams total of both solid and fibrous particles collected shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:12Z
Check the source to verify that this is the current version before use.

4.6.23.4 Particulate matter collection filter. The particulate matter collection filter shall be a multilayer, stainless steel wirewound or stainless steel wire mesh cloth type in which the wires have been fused under controlled conditions and which has a thickness that will insure the specified performance. The filter shall be capable of passing not less than 125 scfm of air or nitrogen to a discharge pressure of atmospheric when used as specified in 4.6.24.3, and shall remove 98 percent by weight of all particles whose two smallest dimensions are 10 microns or greater (10 micron nominal rating). The filter shall further remove all particles whose two smallest dimensions are 40 microns or greater (40 micron absolute rating). The filter element shall be fused or otherwise fastened to the housing in a manner that will insure against gas bypassing the filter element.

4.7 Inspection of preparation for delivery. Preservation, packaging, and marking shall be inspected to verify conformance to the requirements of section 5 herein, and the contract.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Cleaning. The trailer shall be thoroughly cleaned to remove excess and spilled lubrication materials, loose or chipped paint, spilled chemicals, and other foreign materials. All cleaning solvents shall be thoroughly removed from the trailer components prior to delivery.

5.1.1.1 Storage cylinders. The inside of each individual storage cylinder shall be thoroughly cleaned by shot blasting, or an equally effective cleaning process, to provide a uniformly clean surface free of rust, slag, scale, flux or other foreign materials.

5.1.1.2 Degreasing. The storage cylinder, manifold, valve, fitting and other trailer surfaces that will be in contact with the gas being transported, stored, or serviced shall be degreased by one of the following methods. Precautions shall be taken to insure that solvents do not contact parts fabricated from incompatible materials.

Method A - Vapor degreasing process, using stabilized trichloroethylene conforming to MIL-T-7003 in a standard commercial vapor degreaser or by blowing degreasing vapors into the component in such a manner that the vapor will condense on and properly clean all surfaces requiring degreasing. Operation of a commercial vapor degreaser shall be in accordance with the manufacturer's recommendations. Following the vapor degreasing treatment, all solvent shall be removed by baking in an oven, by purging with hot, dry, oil-free air or nitrogen, or by vacuum evacuation.

Method B - Solvent degreasing process, using stabilized trichloroethylene or commercial oxygen-safe cleaning solvent at ambient temperatures to thoroughly wash all surfaces requiring degreasing. If the solvent contains

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:12Z
Check the source to verify that this is the current version before use.

more than 1 percent oil or hydrocarbons after the cleaning and being drained from the component, the degreasing operation shall be repeated with clean solvent. Following the degreasing process, all solvent shall be removed by baking in an oven; by purging with hot, dry, oil-free air or nitrogen, or by vacuum evacuation.

Method C - Detergent degreasing process, in which the components to be cleaned are washed with hot inhibited alkaline cleaner until free from oil, grease, and other contaminant materials. Following this treatment all surfaces (internal and external) shall be rinsed thoroughly with fresh, clean, hot water and dried by blowing with filtered and dried oil-free air or nitrogen, or by baking at a temperature of 250° to 300° F until all wear is removed.

5.1.1.2.1 Petroleum and other flammable solvents shall not be used on such surfaces.

5.1.2 Storage cylinder, manifold and servicing system preparation. Following the cleaning, pressurizing and testing specified herein, the trailer shall be pressurized to not less than 75 psi with clean, dry air or nitrogen. The air or nitrogen shall contain not more than 25 parts per million water vapor by volume and not more than 1 part per million by volume of total hydrocarbons expressed as methene equivalent. With the cartridge removed from the purifier cylinder and with all valves open except the recharging valve, discharge the trailer to a pressure of 30 ± 5 pounds through the servicing system. All valves shall then be tightly closed and remain so to retain the gas. All protective caps shall be securely installed on their respective fittings, after which the ends of all pipes, tubes, couplings, and fittings shall be securely sealed with pressure sensitive tape conforming to PPP-T-60. Type A tags conforming to UU-T-81, with tag and printing waterproofed, stating that the storage cylinders, manifold assembly and servicing system are clean and pressurized with dry, clean, oil free gas and the cartridge has been removed from the purifier cylinder shall be securely attached on or near the servicing, manifold, recharging and cylinder bank valves and the purifier cylinder and access panels in a conspicuous location.

5.1.3 Repainting. Trailer surfaces on which the paint is damaged or defective shall be thoroughly cleaned and repainted as specified under 3.11.

5.1.4 Lubrication. All trailer machined bearing surfaces that are normally lubricated in operation, that do not come into contact with the product carried, and for which preservation requirements are not specified herein shall be thoroughly lubricated with the recommended lubricants to be ready for immediate operation.

5.1.4.1 Preservatives. Preservatives and lubricants shall not be applied to any part of the equipment that will come into contact with the product carried.

5.1.5 Securing for shipment. Small items and packages not fastened to major components shall be secured to or packed with major components to insure against accidental loss or damage caused by movement during shipping.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:12Z
Check the source to verify that this is the current version before use.

5.2 Packing.

5.2.1 Level A. The trailer shall be packed in a crate conforming to PPP-C-650. Cushioning, blocking, bracing and waterproofing shall be in accordance with MIL-STD-1186.

5.2.2 Level C. Packages that require overpacking for acceptance by the carrier shall be packed in exterior shipping containers in a manner that will insure safe transportation, at the lowest rate, to the point of delivery. Containers shall meet Consolidated Freight Classification rules or regulations of other common carriers as applicable to the mode of transportation.

5.3 Marking for shipment. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

TRAILER, COMPRESSED GAS CYLINDER, AIR OR NITROGEN
OIL PUMPED
SERVICING, 2-WHEEL, 8-CYLINDER, 3500 PSI
TYPE MD-3A

6. NOTES

6.1 Intended use. The Type MD-3A trailer is intended for use in servicing aircraft and missile air or nitrogen systems, both high (3,000 psi) and low (25 psi) pressure.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. That MIL-M-008090D(USAF), dated 21 Feb 61, or later issue should be used in conjunction with this specification (see 3.7.7).

c. If the main wheels of the trailer are to be furnished without fenders (see 3.8.1).

d. Registration numbers required (see 3.12.6).

e. Point of preproduction testing (see 4.4).

f. Level of preservation, packaging and packing required (see section 5).

6.3 Definitions. For the purpose of this specification, the following definitions will apply:

6.3.1 Reliability. Reliability is defined as the probability of performing a specified function under given conditions without failure for a specified period of time. Recognizing that, in general, the rate of failure of equipment is fairly constant throughout the life of the equipment, the probability

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:12Z
Check the source to verify that this is the current version before use.

of nonfailure over an operating time interval decreases exponentially as a function of the length of the interval, during which time there is a constant failure rate, and can be expressed as follows:

Confidence level: 90 percent

$$\text{MTBF} = \frac{\text{Total test time}}{2.3}$$

$$\text{Reliability} = e^{-x} \text{ where } x = \frac{\text{Mission Time}}{\text{MTBF}}$$

The $2.3 = \frac{4.61}{2}$ is based on the constant for the Poisson/Chi squared distribution even through the failure rate for the test time is zero.

6.3.2 Mean-time-between-failures. The mean life or mean-time-between-failures is the arithmetical mean (average) of the operating time between failures.

6.3.3 Pressures. Unless specified as absolute or denoted by the symbol psia, all pressures referred to herein will be interpreted as pounds per square inch (psi) gage.

6.3.4 Ambient conditions. Ambient conditions are the dry-bulb temperature, wet-bulb temperature, and the relative humidity of the atmospheric air surrounding and in the vicinity of the trailer, but unaffected by any heat or cold emanating from the trailer itself.

6.3.5 Cycle of operation. A cycle of operation will be considered as opening from 30 pound inches torque closure under operating pressure to the half opening position and return to the 30 pound inches torque closure position.

6.3.6 Fibrous particle. A fibrous particle is defined as a long, slender particle whose maximum cross-sectional dimension is 40 microns.

6.3.7 Standard gas. Standard gas is defined as gas at 29.92 inches Hg absolute and 70° F.

6.4 Certain provisions of 3.7.1 of this specification are the subject of international standardization agreements ASCC Standards 14/6D and 14/7D. When amendment, revision or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international standardization agreement.

International interest (see Section 6).

Custodian:
Air Force - 84

Preparing Activity:
Air Force - 84

Project Number:
2330-F500

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:12Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID

U.S.MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

San Antonio ALC (MMSS)
Kelly AFB TX 78241

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:12Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

☆ U.S. GOVERNMENT PRINTING OFFICE: 1973—714-566/3983 2-1

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:12Z
Check the source to verify that this is the current version before use.